

Surname: _____

Initials: _____

St No. _____

(1) Print on A4 paper. (2) While you are preparing for the lab session, refer to the laboratory manual to complete the preparation sheet in your own handwriting in coloured ink – no black ink or ordinary pencil may be used. (3) If a heading is not applicable to your experiment, leave it open. (4) You have to leave the sheet at the completed experiment in your report book. (5) If you do not hand in your preparation, you will lose two marks. (6) You will earn either 1 or 2 **bonus marks** depending on the quality of your preparation. (7) Copying of someone else's work will be viewed in a serious light.

1. Practical abbreviation and full nameSHM: Simple Harmonic ~~AK~~ Motion**2. Goal**

We will investigate the oscillations of a mass piece hanging on a spring to determine if the mass piece exhibits simple harmonic motions.

3. Safety precautions

~~NI/NA~~ supporting rod on the laboratory stand could poke you in the eye.
spring could shoot you in the eye if it comes undone.

4. Basic equations used in the practical

$$F_R = k y \quad \theta = \omega t \quad \omega = \frac{2\pi}{T} \text{ rad/s}$$

5. Data to be collected

restoring force of spring F_R time period, T of the oscillations.

(1)

6. Table headings including Data collected and calculated data. Mention units of each measurement.

Reading	Force (N)	displacement (mm)	k , spring constant (N/mm)	Reading	Time period, T (s)	Force, F_R (N)	y-displacement, y_e (mm)

7. Sketch the axes of the graph(s) to be drawn with axis labels and units. What type of graph paper will be used?

Rough sketch sketch

8. What data will be obtained from the graph? Include calculation formula (including units).

Time period of oscillations

$$T_{\text{avg}} = \frac{T_1 + T_2 + \dots + T_n}{n}$$